

# Chapter 3, Section 5

James Lee

April 24, 2026

1. Let  $X$  be a projective scheme over a field  $k$ , and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . We define the *Euler characteristic* of  $\mathcal{F}$  by

$$\chi(\mathcal{F}) = \sum (-1)^i \dim_k H^i(X, \mathcal{F}).$$

If

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is a short exact sequence of coherent sheaves on  $X$ , show that  $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$ .

*Proof.* We get an exact sequence of finite dimensional  $k$ -vector spaces (5.2)

$$\dots \xrightarrow{\delta_{i-1}} H^i(X, \mathcal{F}') \longrightarrow H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F}'') \xrightarrow{\delta_i} \dots,$$

where the  $\delta_i$ 's are connecting homomorphisms. By exactness, we have

$$\begin{aligned} \chi(\mathcal{F}) &= \sum (-1)^i \dim_k H^i(X, \mathcal{F}) \\ &= \sum (-1)^i (\dim_k \operatorname{coker} \delta_{i-1} + \dim_k \ker \delta_i) \\ &= \sum (-1)^i (\dim_k H^i(X, \mathcal{F}') - \dim_k \operatorname{im} \delta_{i-1} + \dim_k H^i(X, \mathcal{F}'') - \dim_k \operatorname{im} \delta_i) \\ &= \sum (-1)^i \dim_k H^i(X, \mathcal{F}') + \sum (-1)^i \dim_k H^i(X, \mathcal{F}'') \\ &= \chi(\mathcal{F}') + \chi(\mathcal{F}''). \end{aligned}$$

□

2. (a) Let  $X$  be a projective scheme over a field  $k$ , let  $\mathcal{O}_X(1)$  be a very ample invertible sheaf on  $X$  over  $k$ , and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . Show that there is a polynomial  $P(z) \in \mathbb{Q}[z]$ , such that  $\chi(\mathcal{F}(n)) = P(n)$  for all  $n \in \mathbb{Z}$ . We call  $P$  the *Hilbert polynomial* of  $\mathcal{F}$  with respect to the sheaf  $\mathcal{O}_X(1)$ .
- (b) Now let  $X = \mathbb{P}_k^r$ , and let  $M = \Gamma_*(\mathcal{F})$ , considered as a graded  $S = k[x_0, \dots, x_r]$ -module. Use (5.2) to show that the Hilbert polynomial of  $\mathcal{F}$  just defined is the same as the Hilbert polynomial of  $M$  defined in (I, §7).

*Proof.*

- (a) Since  $\mathcal{O}_X(1)$  is a very ample sheaf on  $X$  over  $k$ , there is a closed immersion  $i : X \rightarrow \mathbb{P}_k^r$  of schemes over  $k$ , for some  $r$ , such that  $\mathcal{O}_X(1) = i^* \mathcal{O}_{\mathbb{P}^r}(1)$ . If  $\mathcal{F}$  is coherent on  $X$ , then  $i_* \mathcal{F}$  is coherent on  $\mathbb{P}_k^r$  (II, Ex. 5.5), and the cohomology is the same (2.10). The Euler characteristic is defined in terms of cohomology, and the projection formula (II, Ex. 5.1d) gives us

$$i_*(\mathcal{F}(n)) = i_*(\mathcal{F} \otimes_{\mathcal{O}_X} i^* \mathcal{O}_{\mathbb{P}^r}(n)) \cong i_* \mathcal{F} \otimes_{\mathcal{O}_{\mathbb{P}^r}} \mathcal{O}_{\mathbb{P}^r}(n) = (i_* \mathcal{F})(n).$$

Thus, we reduce to the case  $X = \mathbb{P}_k^r$ .

Cover  $X$  by a finite number of open affines  $U_i \cong \operatorname{Spec} k[y_1, \dots, y_r]$ . Then there exist finitely generated  $k[y_1, \dots, y_r]$ -modules  $M_i$  such that  $\mathcal{F}|_{U_i} \cong \tilde{M}_i$ . Since  $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F}$  is a closed subset of  $\operatorname{Spec} A_i$ ; in fact,  $\operatorname{Spec} A_i \cap \operatorname{Supp} \mathcal{F} = V(\operatorname{Ann} M_i)$  (AM, Ex. 3.19),  $\operatorname{Supp} \mathcal{F}$  is a closed subset of  $X$ . We proceed by induction on  $\dim \operatorname{Supp} \mathcal{F}$ . If  $\operatorname{Supp} \mathcal{F}$  has dimension zero, then  $\operatorname{Supp} \mathcal{F} \cap U_i$  is affine (II, Ex. 3.11b) equal to spectra of a Noetherian ring  $A$  of dimension zero, which are precisely Artinian rings (AM, 8.5). So  $U_i \cap \operatorname{Supp} \mathcal{F}$  is a finite set (AM, Ex. 8.2), and since  $X$  can be covered by a finite number of  $U_i$ ,  $\operatorname{Supp} \mathcal{F}$  is a finite set of closed points. Thus,  $\mathcal{F}$  is isomorphic to a direct sum of skyscraper sheaves. More precisely,  $\mathcal{F} = \bigoplus_{P \in \operatorname{Supp} \mathcal{F}} \mathcal{F}_P$ , where  $\mathcal{F}_P$  is the stalk of  $\mathcal{F}$  at  $P$ , and cohomology commutes with direct sums, so by (Ex. 1), we reduce to the case when  $\operatorname{Supp} \mathcal{F}$  consists of a single point. Then,  $\chi(\mathcal{F}(n))$  is just the Hilbert polynomial of  $\mathcal{F}_P$  (I, 7.5) as a graded  $A[y_1, \dots, y_r]$ -module.

Assume the statement to be true for all coherent sheaves with support dimension less than  $n$ . Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{x_r} \mathcal{O}_X \longrightarrow \mathcal{O}_H \longrightarrow 0,$$

where  $H$  is the hyperplane  $x_r = 0$ . Taking the tensor product of this sequence with  $\mathcal{F}$  gives us

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0,$$

where  $\mathcal{R}$  is the kernel of the map  $\mathcal{F}(-1) \rightarrow \mathcal{F}$ . Twisting by  $\mathcal{O}_X(n)$  is exact, so we get

$$\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = \chi(\mathcal{F}_H(n)) - \chi(\mathcal{R}(n))$$

by (III, Ex. 5.1). Consider the coherent  $\mathcal{O}_X$ -module  $\mathcal{R}$ . It vanishes under the multiplication map  $x_r$ , where  $x_r$  locally defines one of the coordinates such that  $x_r = 0$  is the defining equation for  $H$ , and such that there is an inclusion  $\text{Supp } \mathcal{R} \subseteq H$ , which exists by Bertini's theorem. Since  $H \cong \mathbf{P}_k^{r-1}$ , we can apply the induction hypothesis to  $\mathcal{R}$  and  $\mathcal{F}_H$ , so let  $P_{\mathcal{R}}(z), P_{\mathcal{F}_H}(z) \in \mathbf{Q}[z]$  be polynomials such that

$$\chi(\mathcal{R}(n)) = P_{\mathcal{R}}(n), \quad \chi(\mathcal{F}_H(n)) = P_{\mathcal{F}_H}(n).$$

The difference function of  $\chi(\mathcal{F}(n))$  is equal to  $P_{\mathcal{F}_H}(z) - P_{\mathcal{R}}(z) \in \mathbf{Q}[z]$ , so by (§1, 7.3), we conclude that  $\chi(\mathcal{F}(n)) = P(n)$  for some  $P(z) \in \mathbf{Q}[z]$ .

- (b) By (5.2), there exists an integer  $n_0$  such that for each  $i > 0$  and each  $n \geq n_0$ ,  $H^i(X, \mathcal{F}(n)) = 0$ . Let  $P(n)$  be the Hilbert polynomial just defined, and let  $P'(n)$  be the polynomial from associated to the graded  $S$ -module  $M$  from (I, §7). Then  $P(n) = \dim_k H^0(X, \mathcal{F}(n))$  for each  $n \geq n_0$ , but  $M_n = \Gamma(X, \mathcal{F}(n)) = H^0(X, \mathcal{F}(n))$  and  $P'(n) = \dim_k M_n$ ; in particular,  $P(n)$  and  $P'(n)$  agree at infinitely many values. Hence, they must be equal. □

**3. Arithmetic Genus.** Let  $X$  be a projective scheme of dimension  $r$  over a field  $k$ . We define the *arithmetic genus*  $p_a$  of  $X$  by

$$p_a(X) = (-1)^r (\chi(\mathcal{O}_X) - 1).$$

Note that it depends only on  $X$ , not on any projective embedding.

- (a) If  $X$  is integral, and  $k$  algebraically closed, show that  $H^0(X, \mathcal{O}_X) \cong k$ , so that

$$p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k H^{r-i}(X, \mathcal{O}_X).$$

In particular, if  $X$  is a curve, we have

$$p_a(X) = \dim_k H^1(X, \mathcal{O}_X).$$

- (b) If  $X$  is a closed subvariety of  $\mathbf{P}_k^r$ , show that this  $p_a(X)$  coincides with the one defined in (I, Ex. 7.2), which apparently depended on the projective embedding.
- (c) If  $X$  is a nonsingular projective curve over an algebraically closed field  $k$ , show that  $p_a(X)$  is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree  $d \geq 3$  is not rational.

*Proof.*

- (a) For any embedding  $j : X \rightarrow \mathbf{P}_k^r$ ,  $H^0(X, \mathcal{O}_X) \cong k$  (I, 3.4).
- (b) Let  $\mathcal{I}_X$  be the ideal sheaf of  $X$  on  $\mathbf{P}_k^r$ . The homogenous coordinate ring  $S(X)$  of  $X$  for the given embedding is  $k[x_0, \dots, x_r]/I$ , where  $I$  is the ideal  $\Gamma_*(\mathcal{I}_X)$  (II, Ex. 5.14). The result follows by (Ex. 1) and (Ex. 2b).
- (c) Two nonsingular projective curves are birational if and only if they are isomorphic (II, 6.7), so they must have the same cohomology. If  $X$  is a nonsingular plane curve of degree  $d \geq 3$ , then

$$p_a(X) = \frac{1}{2}(d-1)(d-2) > 0$$

by (I, Ex. 7.2), but  $p_a(\mathbf{P}^1) = 0$ . □

**7.** Let  $X$  (respectively,  $Y$ ) be proper schemes over a Noetherian ring  $A$ . We note by  $\mathcal{L}$  an invertible sheaf.

- (a) If  $\mathcal{L}$  is ample on  $X$ , and  $Y$  is any closed subscheme of  $X$ , then  $i^*\mathcal{L}$  is ample on  $Y$ , where  $i : Y \rightarrow X$  is the inclusion.
- (b)  $\mathcal{L}$  is ample on  $X$  if and only if  $\mathcal{L}_{\text{red}} = \mathcal{L} \otimes \mathcal{O}_{X_{\text{red}}}$  is ample on  $X_{\text{red}}$ .
- (c) Suppose  $X$  is reduced. Then  $\mathcal{L}$  is ample on  $X$  if and only if  $\mathcal{L} \otimes \mathcal{O}_{X_i}$  is ample on  $X_i$ , for each irreducible component  $X_i$  of  $X$ .

- (d) Let  $f : X \rightarrow Y$  be a finite surjective morphism, and let  $\mathcal{L}$  be an invertible sheaf on  $Y$ . Then  $\mathcal{L}$  is ample on  $Y$  if and only if  $f^*\mathcal{L}$  is ample on  $X$ .

*Proof.*

- (a) If  $\mathcal{F}$  is any coherent sheaf on  $Y$ , then  $f_*\mathcal{F}$  is a coherent  $\mathcal{O}_X$ -module (II, 5.5c)  
 (b)  
 (c)  
 (d) By (b) and (c), we reduce to the case  $X$  and  $Y$  are irreducible, reduced, and proper schemes over  $A$ .

□

10. Let  $X$  be a projective scheme over a Noetherian ring  $A$ , and let  $\mathcal{F}^1 \rightarrow \mathcal{F}^2 \rightarrow \dots \rightarrow \mathcal{F}^r$  be an exact sequence of coherent sheaves on  $X$ . Show that there is an integer  $n_0$ , such that for all  $n \geq n_0$ , the sequence of global sections

$$\Gamma(X, \mathcal{F}^1(n)) \longrightarrow \Gamma(X, \mathcal{F}^2(n)) \longrightarrow \dots \longrightarrow \Gamma(X, \mathcal{F}^r(n))$$

is exact.

*Proof.* Denote  $f_i : \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}$  for each  $i = 1, \dots, r-1$ . For every  $i = 2, \dots, r-1$ , we have the following three short exact sequence

$$0 \longrightarrow \operatorname{im} f_{i-1} \longrightarrow \mathcal{F}^i \longrightarrow \operatorname{im} f_i \longrightarrow 0,$$

$$0 \longrightarrow \ker f_{i-1} \longrightarrow \mathcal{F}^{i-1} \longrightarrow \operatorname{im} f_{i-1} \longrightarrow 0,$$

$$0 \longrightarrow \operatorname{im} f_i \longrightarrow \mathcal{F}^{i+1} \longrightarrow \operatorname{coker} f_i \longrightarrow 0.$$

Every sheaf above is coherent (II, 5.7). By (5.2), there exists an integer  $n_i$  such that for each  $i > 0$  and each  $n \geq n_i$ , the cohomology vanishes for every sheaf, i.e., there is an exact sequence of global sections

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \ker f_{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \operatorname{im} f_{i-1}(n)) \longrightarrow 0$$

$$0 \longrightarrow \Gamma(X, \operatorname{im} f_i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n)) \longrightarrow \Gamma(X, \operatorname{coker} f_i(n)) \longrightarrow 0$$

which implies the sequence

$$\Gamma(X, \mathcal{F}^{i-1}(n)) \longrightarrow \Gamma(X, \mathcal{F}^i(n)) \longrightarrow \Gamma(X, \mathcal{F}^{i+1}(n))$$

is exact. Take  $n_0 = \max n_i$  to obtain the desired integer.

□